Circles & Coordinate Geometry

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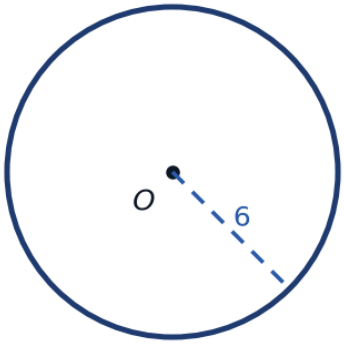
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The equation of a circle

Write the equation of a circle with center $(-4, 7)$ and radius 9.

2 [Foundational] • Target: The equation of a circle

The circle below has the radius shown. Find its area and circumference, in terms of π.



3 [Foundational] • Target: The equation of a circle

A circle has diameter 26 and is centered at the origin. Write its equation, and state its radius.

4 [Foundational] • Target: Completing the square

Find the center and radius of $x^2 + y^2 - 6x + 8y - 11 = 0$.

5 [Foundational] • Target: Distance, midpoint, and slope

Find the distance between $(-3, 5)$ and $(4, -7)$, to the nearest hundredth.

6 [Foundational] • Target: Distance, midpoint, and slope

Find the midpoint and slope of the segment joining $(2, -6)$ and $(-8, 10)$.

7 [Foundational] • Target: Distance, midpoint, and slope

Find the slope of the line through $(-2, 7)$ and $(6, -5)$, and state whether the line rises or falls from left to right.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Coordinate proofs**

Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle, using slopes.

9 **[Intermediate]** • **Target: Coordinate proofs**

Show that $A(1, 2)$, $B(4, 3)$, $C(3, 6)$, $D(0, 5)$ form a parallelogram, by showing both pairs of opposite sides are parallel.

10 **[Intermediate]** • **Target: Tangent lines to circles**

A circle has center $O(2, 3)$ and radius 5. Point $P(2, 12)$ lies outside the circle. Find the length of the tangent from P to the circle, to the nearest hundredth.

11 **[Intermediate]** • **Target: Tangent lines to circles**

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$ on the circle.

12 **[Intermediate]** • **Target: Tangent lines to circles**

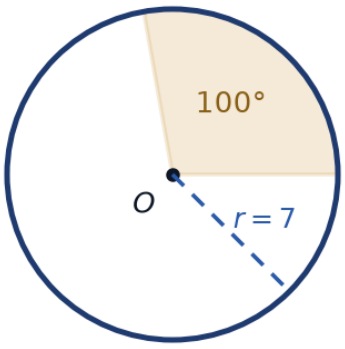
A circle has center $(0, 0)$ and radius 10. Determine whether the point $(6, 9)$ lies inside, on, or outside the circle.

13 **[Intermediate]** • **Target: Circle theorems**

A central angle of 84° subtends an arc. Find the measure of the inscribed angle subtending the same arc.

14 **[Intermediate]** • **Target: Circle theorems**

A chord subtends the central angle shown below. Find the measure of the inscribed angle subtending the same (minor) arc.



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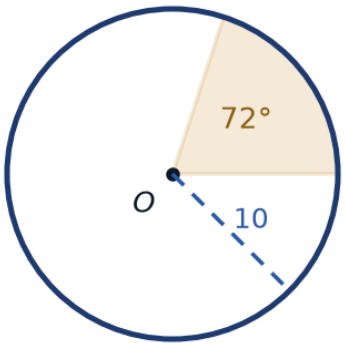
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Arc length and sector area • Cross-Lesson: 7.01 Solid Geometry & Similarity

A circle has radius 12 cm. Find the arc length subtended by a central angle of $\frac{5\pi}{6}$ radians.

16 [Challenging] • Target: Arc length and sector area • Cross-Lesson: 8.01 Statistics & Probability

The circle below has the radius and shaded sector shown. Find the area of the sector, in terms of π.



17 [Challenging] • Target: Arc length and sector area • Cross-Lesson: 1.01 Advanced Algebra

A sector of a circle with radius 9 has area 27π. Find the central angle, in radians.

18 [Challenging] • Target: Conics: ellipses and hyperbolas • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Identify the type of conic and state its center for $\frac{(x-2)^2}{16}+\frac{(y+1)^2}{9}=1$.

19 [Challenging] • Target: Conics: ellipses and hyperbolas • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Identify the type of conic, its center, and its asymptote slopes for $\frac{(x+3)^2}{4}-\frac{y^2}{9}=1$.

20 [Challenging] • Target: A well-designed word problem: a Ferris wheel • Cross-Lesson: 4.01 Lines, Triangles & Polygons

This question has three parts. A circular Ferris wheel has a diameter of 40 m, with its center 22 m above the ground. Set up a coordinate system with the origin at ground level, directly below the center. (a) Write the equation of the circle traced by a passenger cabin. (b) A support cable runs from a ground anchor at $(26, 0)$ to a point of tangency on the wheel. Find the length of this cable, to the nearest hundredth of a meter. (c) Find the height above the ground of a cabin when it is 12 m horizontally from the vertical line through the center.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The equation of a circle

A circle has center $O(8, 4)$ and radius 20. Point $P(8, 33)$ lies outside the circle. Find the length of the tangent from P to the circle.

a. 29

b. 23

c. 21

d. 19

22 [Foundational] • Target: Completing the square

A circle has radius 8. Find the area of the sector with central angle 60°, in terms of π.

a. $\frac{29}{3}\pi$

b. $\frac{32}{3}\pi$

c. $\frac{64}{3}\pi$

d. $\frac{35}{3}\pi$

23 [Foundational] • Target: Distance, midpoint, and slope

A circle has center $(4, -2)$ and passes through $(13, 10)$. Find the slope of the tangent line to the circle at $(13, 10)$.

a. 5/4

b. 1/4

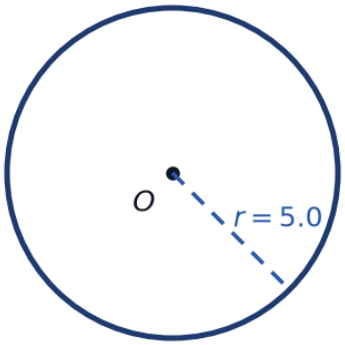
c. -3/4

d. -11/4

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24 [Foundational] • Target: Coordinate proofs

What is the equation of the circle with centre $(-2, 3)$ and radius 4?



a. $(x+2)^2+(y-3)^2=16$

b. $(x-2)^2+(y+3)^2=16$

c. $(x+2)^2+(y-3)^2=4$

d. $(x-2)^2+(y-3)^2=16$

25 [Foundational] • Target: Tangent lines to circles

A circle has center $(6, 6)$ and radius 2. Find the shortest distance from the point $(16, 30)$ to the circle.

a. 6

b. 38

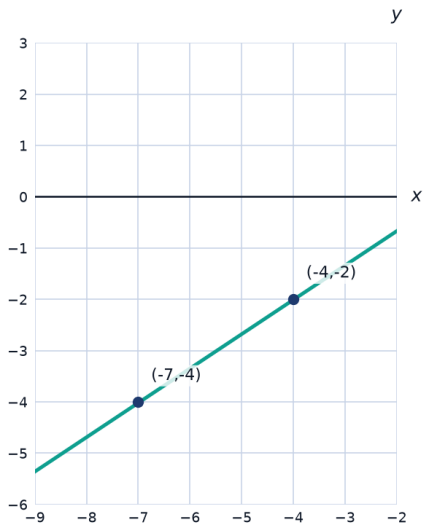
c. 24

d. 29

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26 [Foundational] • Target: Circle theorems

Find the slope of the line through $(-4, -2)$ and $(-7, -4)$.



a. 2/3

b. -4/3

c. 8/3

d. 13/6

27 [Foundational] • Target: Arc length and sector area

The point $(17, y)$ lies on the circle $(x + 3)^2 + (x - 5)^2 = 841$ with $y > 5$. Find y.

a. 26

b. 33

c. 18

d. 25

28 [Foundational] • Target: Conics: ellipses and hyperbolas

Line ℓ_1 has slope -2. What is the slope of a line perpendicular to ℓ_1?

a. 1/2

b. 5/2

c. 1/4

d. 1/6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: a Ferris wheel

A circle is given by $x^2 + y^2 - 2x - 16y + 61 = 0$. If its center is (h, k) and radius r, find $h + k + r$.

a. 11

b. 5

c. 12

d. 7

30 [Intermediate] • Target: The equation of a circle

Find the midpoint of the segment joining $(6, -4)$ and $(8, -10)$.

a. (7,-16)

b. (7,-10)

c. (7,-3)

d. (7,-7)

31 [Intermediate] • Target: Completing the square

Find the distance between $(3, 0)$ and $(-2, -12)$.

a. 11

b. 10

c. 13

d. 20

32 [Intermediate] • Target: Distance, midpoint, and slope

Find the radius of the circle $x^2 + y^2 + 4x + 2y - 11 = 0$.

a. 4

b. 1

c. 9

d. 7

33 [Intermediate] • Target: Coordinate proofs

A circle has center $(-2, 6)$ and passes through $(18, 27)$. Find its radius.

a. 19

b. 17

c. 29

d. 24

34 [Intermediate] • Target: Tangent lines to circles

Triangle has vertices $A(0, 0)$, $B(10, 0)$, $C(0, 24)$. Find the length of the hypotenuse BC.

a. 37

b. 11

c. 26

d. 16

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35 [Intermediate] • Target: Circle theorems

Line ℓ_1 has slope -1. What is the slope of a line parallel to ℓ_1?

a. -7

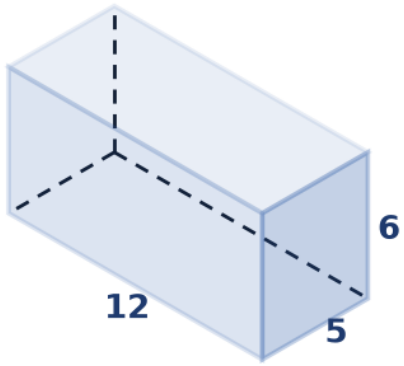
b. 2

c. -1

d. -4

36 [Intermediate] • Target: Arc length and sector area

Find the surface area of a rectangular prism with length 12, width 5, and height 6.



a. 166

b. 324

c. 131

d. 329

37 [Intermediate] • Target: Conics: ellipses and hyperbolas

In the diagram below, lines DE and BC are parallel. If AD = 6, DB = 4, and AE = 9, what is the length of EC?

a. 6

b. 7

c. 8

d. 10

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: A well-designed word problem: a Ferris wheel • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In the circle shown below, chords AB and CD intersect at point P. If $PA = 4$, $PB = 6$, and $PC = 3$, what is the length of segment $\overline{PD}$?

a. 5

b. 7

c. 8

d. 9

39 [Challenging] • Target: The equation of a circle • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Find the surface area of a cube with edge length 9.

a. 377

b. 486

c. 188

d. 49

40 [Challenging] • Target: Completing the square • Cross-Lesson: 4.01 Lines, Triangles & Polygons

$ABCD$ is a parallelogram with $A(4, -6)$, $B(6, -6)$, $C(2, -6)$. Find D.

a. (0,-6)

b. (0,-1)

c. (0,-13)

d. (0,-4)

41 [Challenging] • Target: Distance, midpoint, and slope • Cross-Lesson: 5.01 Trigonometry

Consider a circle with the equation $x^2 + y^2 = 25$. A point $P(x_0, y_0)$ lies on this circle. A tangent line to the circle at point P intersects the x-axis at point A and the y-axis at point B. If the area of the triangle OAB, where O is the origin $(0, 0)$, is minimized, find the value of $x_0^2 + y_0^2$.

a. 25

b. 50

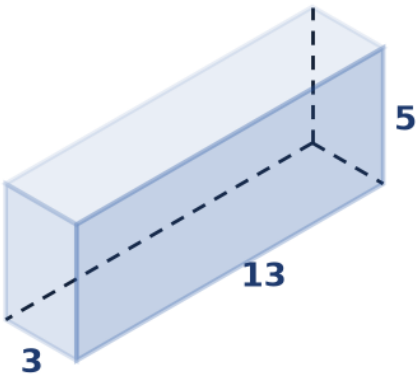
c. 100

d. 125

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42 [Challenging] • Target: Coordinate proofs • Cross-Lesson: 7.01 Solid Geometry & Similarity

Find the volume of a rectangular prism with length 3, width 13, and height 5.



a. 226

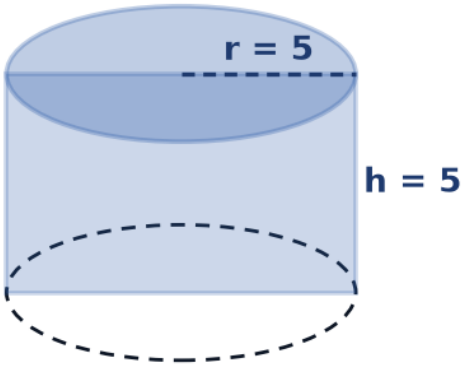
b. 162

c. 183

d. 195

43 [Challenging] • Target: Tangent lines to circles • Cross-Lesson: 8.01 Statistics & Probability

Find the total surface area of a cylinder with radius 5 and height 5. (In terms of π.)



a. 57π

b. 45π

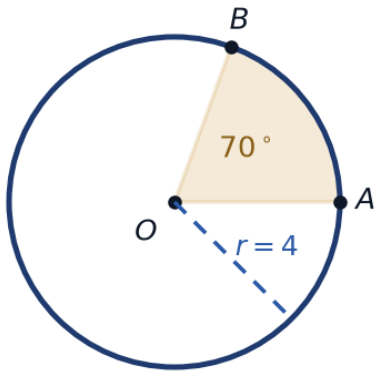
c. 100π

d. 121π

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44 [Challenging] • Target: Circle theorems • Cross-Lesson: 1.01 Advanced Algebra

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.



a. 25

b. 30

c. 35

d. 40

45 [Challenging] • Target: Arc length and sector area • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In the diagram below, points A, B, C, and D lie on the circle. If AB = 6, BP = 8, CD = 4, and AP = x, where P is the intersection of chords AC and BD, find the value of x.

a. 4

b. 5

c. 6

d. 8

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